

Predicting U.S. Bank Distress

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PROJECT TITLE

Predicting U.S. Bank Distress

PROJECT SUMMARY

This project builds an early-warning model that uses public FDIC and FRED quarterly data to flag which U.S. banks are at risk of becoming undercapitalized. The data consists of five linked FDIC tables (1.68M bank-quarter records of each bank's quarterly financial and regulatory filings) plus a FRED macro table (economic context like interest rates, unemployment, and GDP). Data understanding is complete; the next step is feature selection and preparing the data for modeling.

MILESTONES

1. Literature review of bank failure and existing early-warning models (*in progress*)
2. Collect the data (FDIC API, 1984 to present) (**DONE**)
3. Understand the data through exploratory data analysis (**DONE**)
4. Prepare the data for machine learning (*Next*)
5. Train and compare three candidate models

LAST WEEK'S "TO DO"

From the Week 2 report:

1. Continuing the literature review (**in progress**)
2. Understanding the data to do feature selection (**DONE**)

THIS WEEK'S PROGRESS

Completed the data-understanding phase. Consolidated the literature into a features list per dataset, and verified the exact regulatory definitions behind the target label (becoming undercapitalized) against the regulations (12 CFR 324.403 and its predecessors). Built the working modeling table: 1.26M bank-quarter records (1990–2026) joined with bank statics and macro context, with the distress label. The target is rare: 0.73% of healthy bank-quarters cross into distress within the following year.

The key EDA finding (figure below): banks that later became undercapitalized show visible

deterioration across all six core vitals for roughly two years beforehand, while matched healthy banks tracked over the same calendar quarters stay flat. The eventual cases are already measurably weaker two years out. The decline is bank-specific, not only macro-driven.

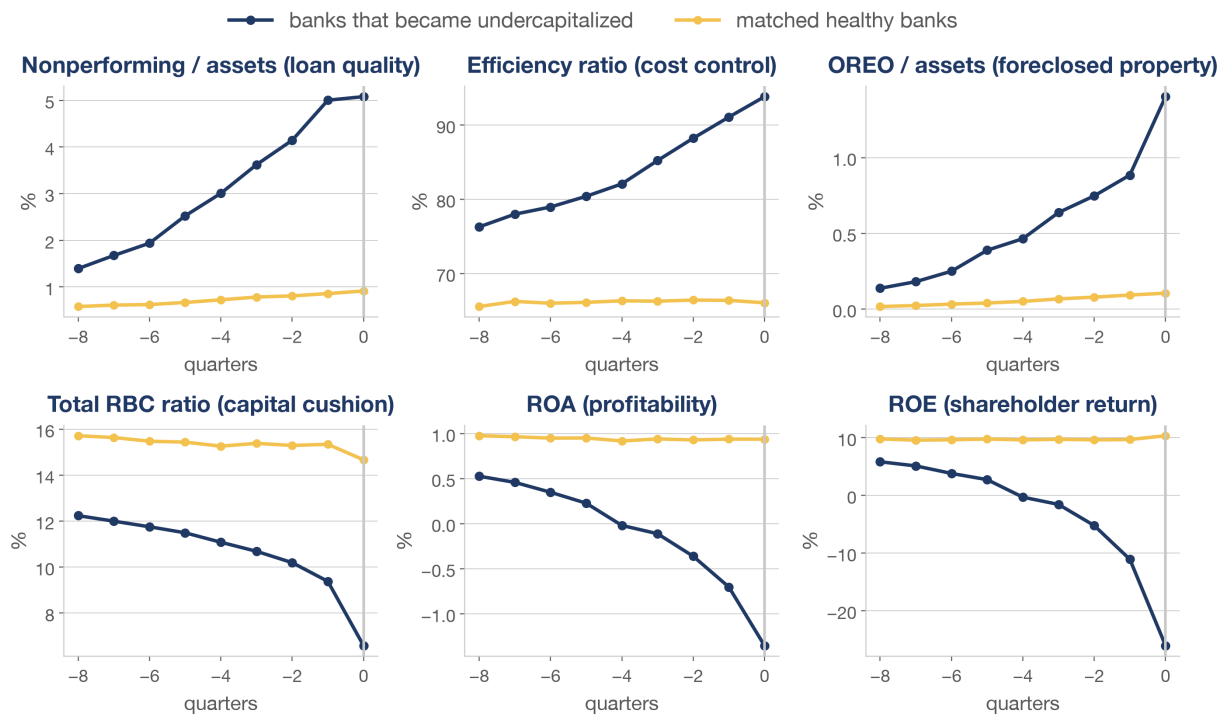


Figure 1

Median financial vitals over the eight quarters before a bank became undercapitalized (quarter 0). Navy: all banks that became undercapitalized, each aligned to its own quarter 0. Gold: healthy banks tracked over the same calendar quarters.

ISSUES & DISCUSSION

- The target is severely imbalanced: about 7 in 1,000 healthy bank-quarters become undercapitalized within a year. The evaluation metric has to be built around catching the rare cases.
- Data leakage is a risk for this problem: the label looks into the future, so training and validation must be split by time, with a gap year, and every feature must use only data available as of the prediction quarter.
- Some failures don't follow the two-year decline pattern: SVB's official ratios looked healthy right up to its collapse. Fast, liquidity-driven failures will need funding-stability features beyond the classic ratios.

NEXT WEEK'S "TO DO"

1. Implement the time-based train/test split
2. Feature selection: filter pass, then model-based ranking

RESOURCES

Literature review sources reviewed so far:

- Cole and White, "Déjà Vu All Over Again: The Causes of U.S. Commercial Bank Failures This Time Around," *Journal of Financial Services Research*, 2012, Vol. 42, No. 1, pp. 5–29.
- Collier, Forbush, Nuxoll, and O'Keefe, "The SCOR System of Off-Site Monitoring: Its Objectives, Functioning, and Performance," *FDIC Banking Review*, 2003, Vol. 15, No. 3.
- Petropoulos, Siakoulis, Stavroulakis, and Vlachogiannakis, "Predicting Bank Insolvencies Using Machine Learning Techniques," *International Journal of Forecasting*, 2020, Vol. 36, No. 3, pp. 1092–1113.
- Correia, Luck, and Verner, "Failing Banks," Federal Reserve Bank of New York Staff Report No. 1117, September 2024, revised June 2025.
- Chu, Frazier, Martin, Agarwal, Kuo, Northwood, Oey, Reed, Rodrigue, and Starr, "Dissecting Depositor Flight: An Analysis of the Spring 2023 Bank Failures," FDIC.
- 12 CFR § 324.403, Capital measures and capital category definitions (the PCA thresholds behind the target label).

Note on LLM use: This document was typeset in LaTeX using Claude Code, which handled the formatting, styling code, and wording polish. All content, research, and decisions are my own, and every suggested edit was reviewed and approved by me.